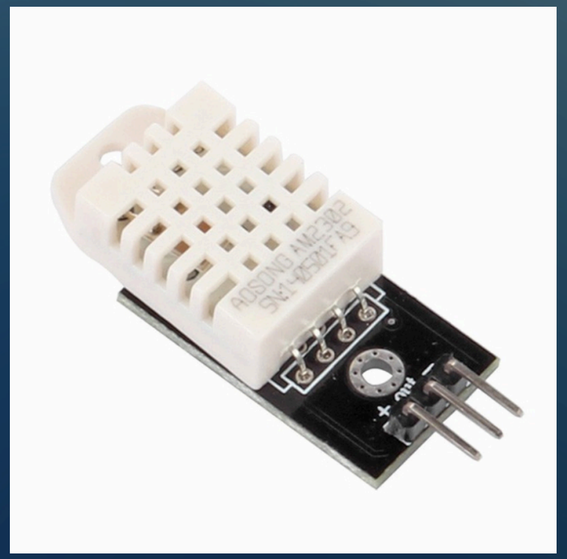
Sensor & Actuator

Focus Discussion Group - Group 1

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Humam Al Labib - 220608175

DHT 22 Module

Temprature and Humidity Sensor



How ?

Data pin send out **Digital Signal** with specific messaging protocol for sending the Temperature and Humidity Value. We can use a library like **Adafruit DHT Sensor Library** that implements the messaging protocol

Notice

We cannot make the sensor to works. We have a validated that its not the board, wiring, and dependencies is not the problematic part.

Specifications	Value
Operating Voltage	3.3v – 5v
Max Current Consumption	2.5mA
Temperature Range	-40 to 80°C
Temperature Accuracy	± 0.5°C
Humidity Range	0 – 100%
Humidity Accuracy	± 2-5%
Sampling Rate	0.5Hz (one reading every 2 seconds)
Body Size	15.1mm x 25mm x 7.7mm

MQ-135

Air Quality sensor



MQ-135

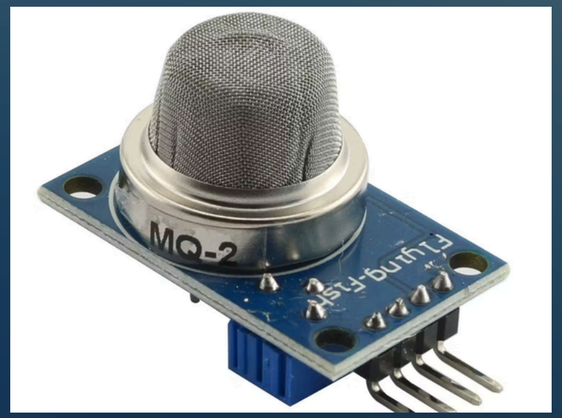
How ?

The data pins send out a raw **Analog Voltage** or a simple **Digital High/Low** signal based on gas levels, requiring no special communication protocols. The analog voltage smoothly rises as the air gets more polluted, while the digital pin acts as a simple on/off trigger switch when a specific gas threshold is crossed. We can tune the threshold by using the on board potentiometer.

Specifications	Value
Circuit Voltage	5V ± 0.1V AC or DC
Heating Voltage	5V ± 0.1V AC or DC
Heater Resistance	33Ω ± 5% (at room temperature)
Heating Power Consumption	Less than 800mW
Operating Temperature Range	-10°C to 45°C
Storage Temperature Range	-20°C to 70°C
Operating Humidity Range	Less than 95% RH
Sensing Resistance	30KΩ – 200KΩ (at 100ppm NH3)
Preheat Time	Over 24 hours
Detecting Concentration Scope	• NH3 : 10ppm – 300ppm • Benzene: 10ppm – 1000ppm • Alcohol: 10ppm – 300ppm

MQ-2

Combustible Gas sensor

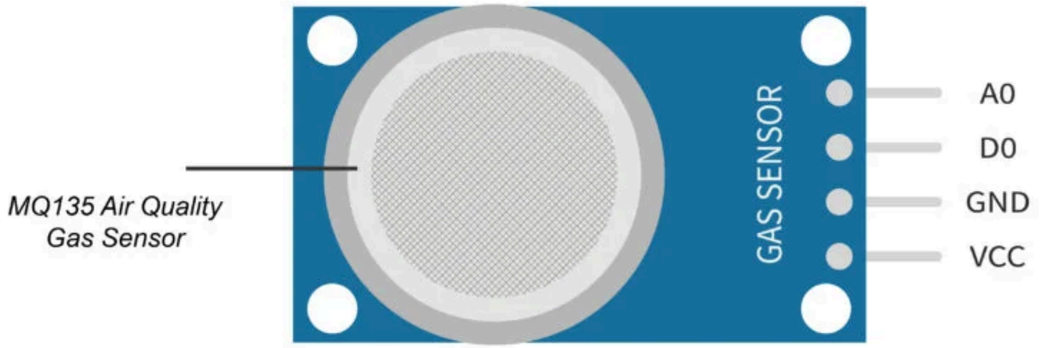


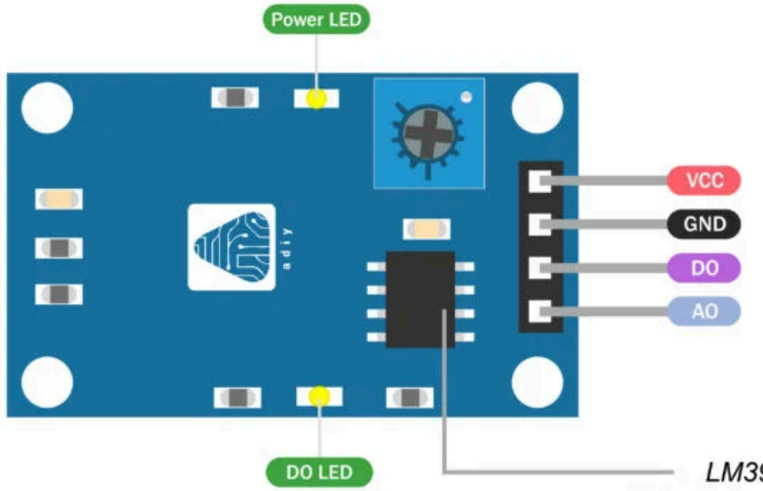
How ?

It works the same as the MQ-2 sensor. The data pins send out a raw **Analog Voltage** or a simple **Digital High/Low** signal based on gas levels, requiring no special communication protocols. The analog voltage smoothly rises as the air gets more polluted, while the digital pin acts as a simple on/off trigger switch when a specific gas threshold is crossed. We can tune the threshold by using the on-board potentiometer.

Specifications	Value
Circuit Voltage	5V ± 0.1V AC or DC
Heating Voltage	5V ± 0.1V AC or DC
Heater Resistance	33Ω ± 5% (at room temperature)
Heating Power Consumption	Less than 800mW
Operating Temperature Range	-20°C to 50°C
Storage Temperature Range	-20°C to 70°C
Operating Humidity Range	Less than 95% RH
Sensing Resistance	3KΩ – 30KΩ (at 1000ppm iso-butane)
Preheat Time	Over 24 hours
Detecting Concentration Scope	• LPG / Propane: 200ppm – 5000ppm • Butane / H2: 300ppm – 5000ppm • Methane: 5000ppm – 20000ppm • Alcohol / Smoke: 100ppm – 2000ppm

ADIY MQ135 Air Quality Gas Sensor Module





HC-SR501

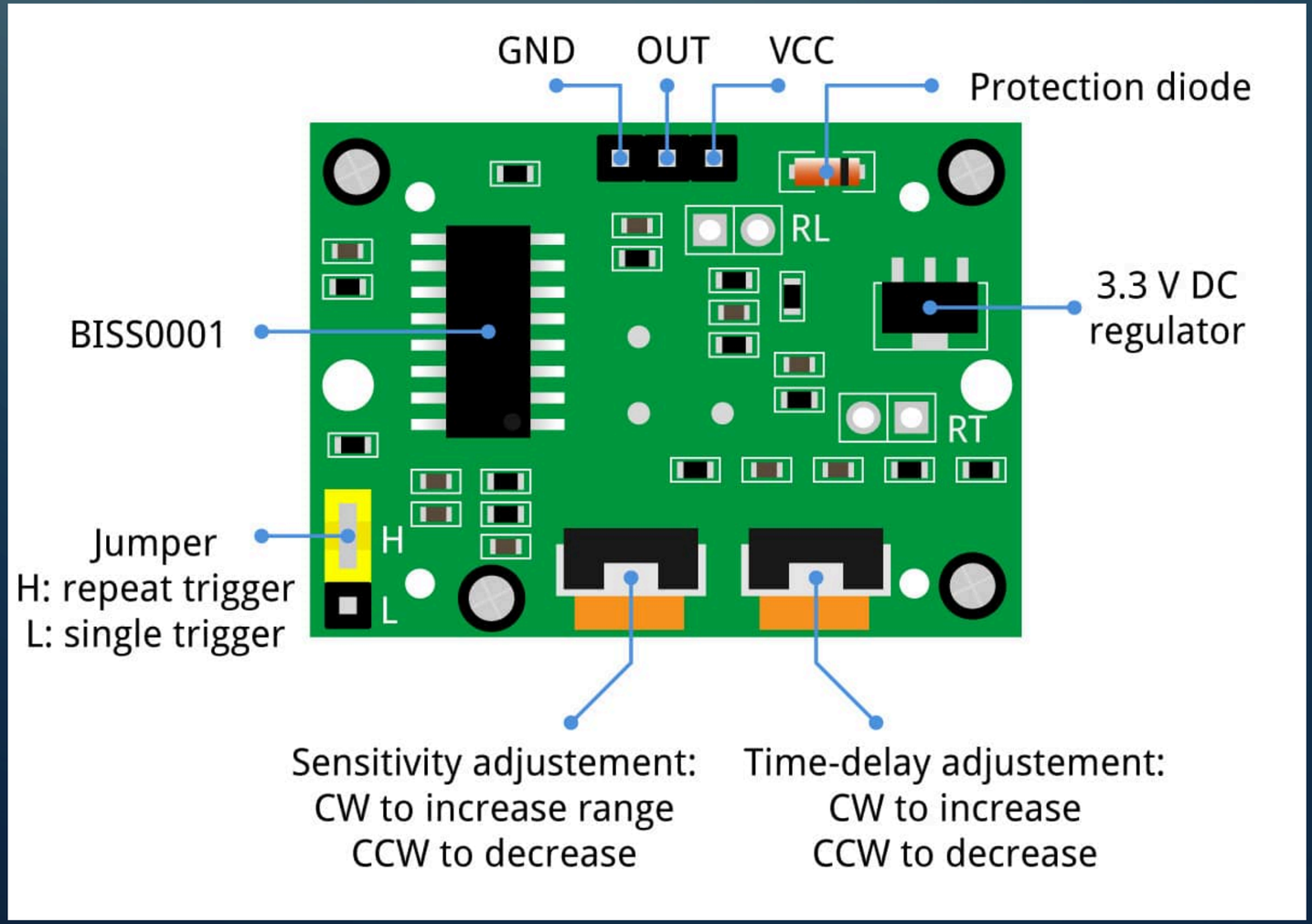
PIR Motion Sensor



How ?

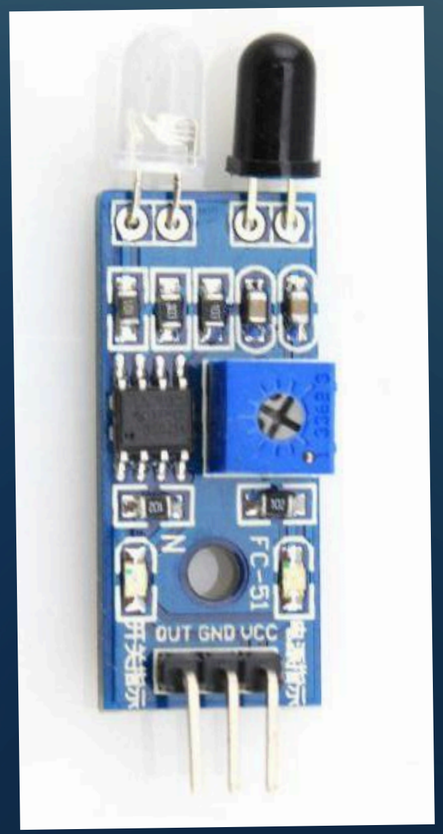
The data pin sends out a simple **Digital High/Low** signal based on infrared heat movement, requiring no special communication protocols. The digital pin acts as a simple on/off trigger switch that jumps to a steady 3.3V High signal when motion is detected, and drops back down to 0V Low when the area becomes still. We can tune the detection distance and the signal duration by using the onboard potentiometers.

Specifications	Value
Operating Voltage Range	DC 4.5V to 20V (5V typical)
Quiescent Current (Standby)	Less than 50µA to 65µA
Level Output (Digital Logic)	HIGH: 3.3V / LOW: 0V (TTL logic)
Sensing Distance Range	3 meters to 7 meters (Adjustable via potentiometer)
Sensing Angle (Cone Spread)	Less than 110° to 120° cone angle
Delay Time Adjustment	~0.3 seconds to 5 minutes (Adjustable via potentiometer)
Block / Lock Time (Default)	2.5 seconds (Standard delay before looking for new motion)
Trigger Methods	• L: Non-repeatable trigger (single pulse) • H: Repeatable trigger (keeps output HIGH while motion continues)
Operating Temperature Range	-15°C to 70°C (up to 80°C on some variants)
Warm-up / Calibration Time	60 seconds to 2 minutes after power-up


GND
OUT
VCC
Protection diode
RL
3.3 V DC
regulator
BISS0001
RT
Jumper
H: repeat trigger
L: single trigger
H
L
Sensitivity adjustement:
CW to increase range
CCW to decrease
Time-delay adjustement:
CW to increase
CCW to decrease

FC-51

IR Proximity sensor



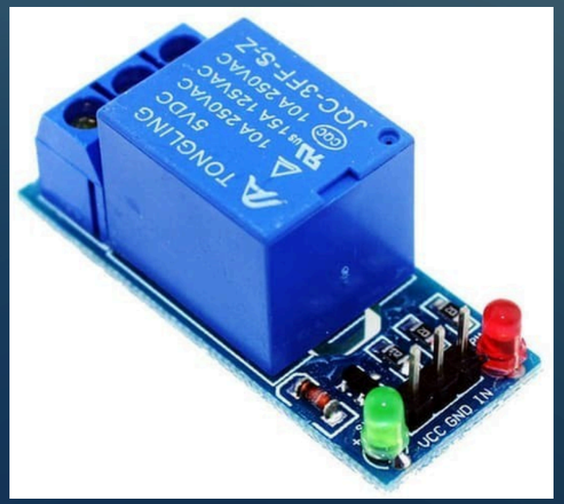
How ?

The data pin sends out a simple **Digital High/Low** signal based on infrared light reflection, requiring no special communication protocols. The digital pin acts as a simple on/off trigger switch that drops to a steady 0V Low signal when an obstacle is detected, and jumps back up to a 5V/3.3V High signal when the path becomes clear. We can tune the detection distance by adjusting the onboard potentiometer, which shifts the trigger threshold of the comparator chip.

Specification	Value
Operating Voltage	3.3V to 5V DC
Output Type	Digital Output (Low when obstacle detected, High when clear)
Detection Range	2 cm to 30 cm (Adjustable via onboard potentiometer)
Detection Angle	~35°
Operating Current	~20 mA
Sensor Elements	1x IR Transmitter (LED), 1x IR Receiver (Photodiode)
Board Indicators	Power LED (Red), Obstacle Detection LED (Green/Red)
PCB Dimensions	Approx. 3.2 cm x 1.4 cm

JQC-3FF-S-Z

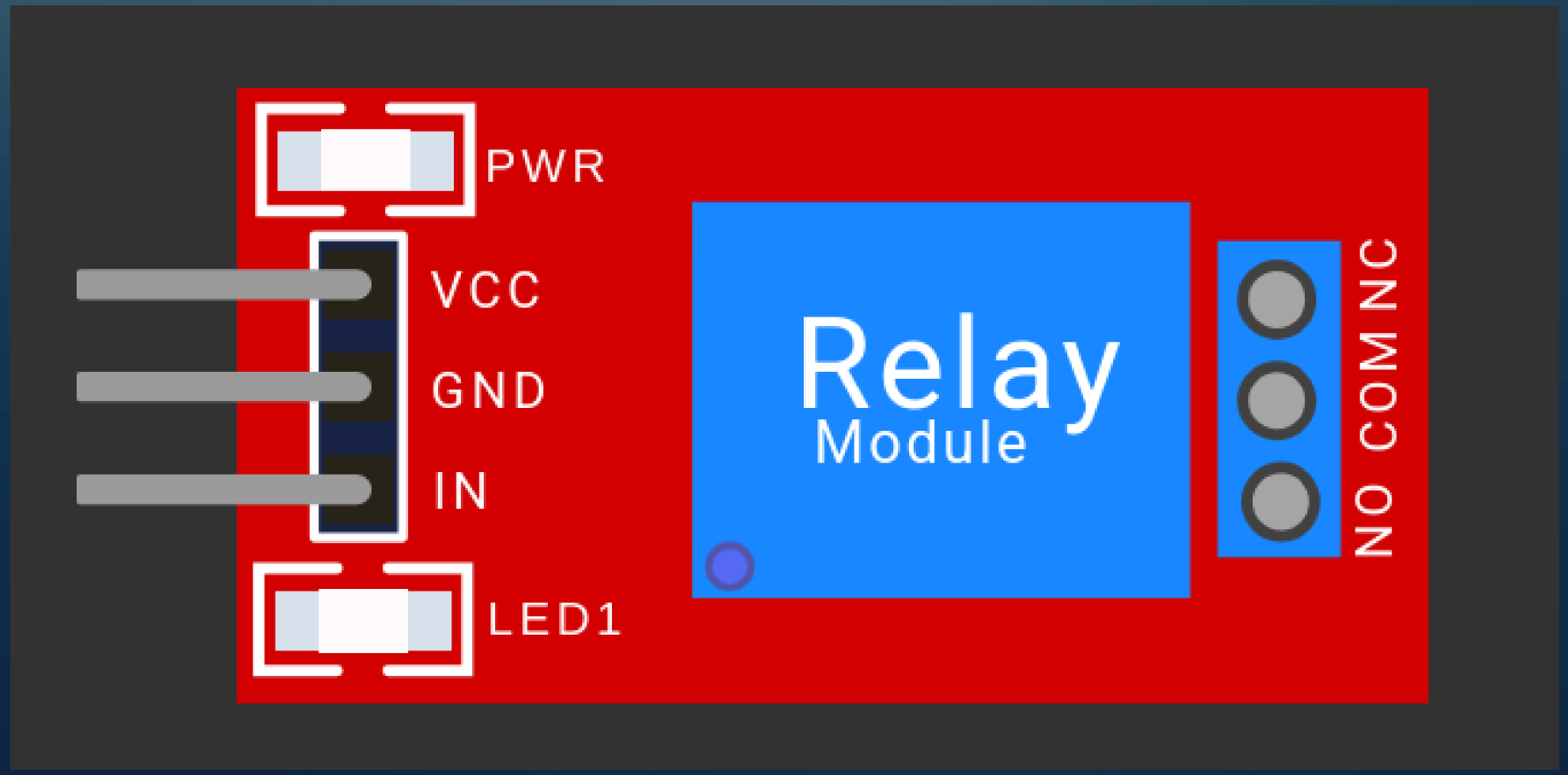
Power Relay



How ?

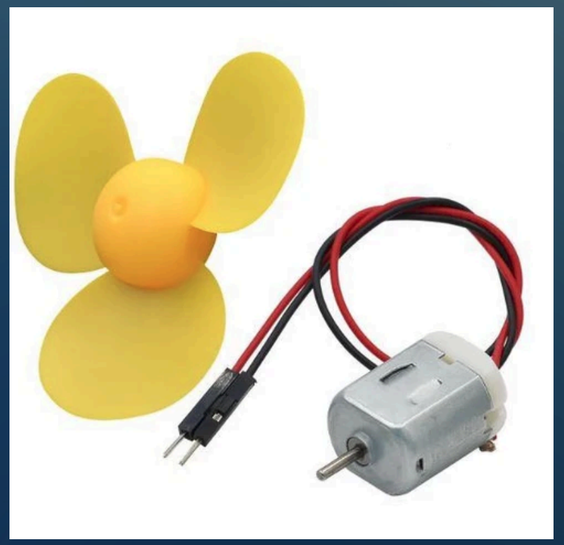
The **control side** activates when a microcontroller sends a simple **Digital High/Low** signal to an external transistor, allowing current to flow through the relay's internal electromagnet coil and create a physical magnetic field. This magnetic field pulls an isolated internal metal arm away from its resting position, acting as an on/off mechanical trigger switch that safely bridges the gap to the separate power side. By shifting this arm, the high-voltage circuit alternates its path between the **Normally Closed (NC)** and **Normally Open (NO)** terminals, completely isolating the low-voltage logic components from dangerous high-power appliances.

Specification	Value
Contact Ratings	10 A at 250 V AC / 15 A at 125 V AC / 10 A at 28 V DC
Max Switching Power	2770 VA (AC) / 240 W (DC)
Max Switching Voltage	250 V AC / 30 V DC
Max Switching Current	15 A
Coil Power	0.36 W
Operate Time	10 ms Max
Release Time	5 ms Max
Insulation Resistance	100 MΩ Min (at 500 V DC)
Dielectric Strength	1500 V AC (Coil to Contact for 1 minute)
Dimensions	19.2 x 15.4 x 15.4 mm
Temperature Range	-40 °C to +85 °C


PWR
VCC
GND
IN
LED1
Relay
Module
NO COM NC

DC Motor / Fan

Electric Actuator



How ?

Only accepts raw electrical power and cannot process digital data signals. To control the shaft's rotations per minute (RPM), an external device must directly regulate the power delivered from the external power source to the motor. Components like a **MOSFET transistor** or a **motor driver module** act as high-speed electronic switches controlled by a microcontroller using **PWM**

Specification	Value
Operating Voltage Range	3V to 12V DC (Rated standard: 6.0V DC)
Starting Voltage	2.0V DC
No-Load Current (Free-Run)	~70 mA (Maximum: 70 mA at 6V)
Normal Loaded Running Current	250 mA (Maximum rated load)
Stall Current (Peak/Locked)	500 mA to 800 mA maximum (at 6V)
No-Load Speed	9,100 to 11,500 RPM (at 6V)
Loaded Speed	4,500 ± 1,500 RPM (under standard load)
Rated Load (Torque)	10 g·cm
Starting Torque	20 g·cm
Operating Temperature Limit	-10°C to +60°C
Motor Unit Weight	~17.5 grams

SG90 Micro Servo

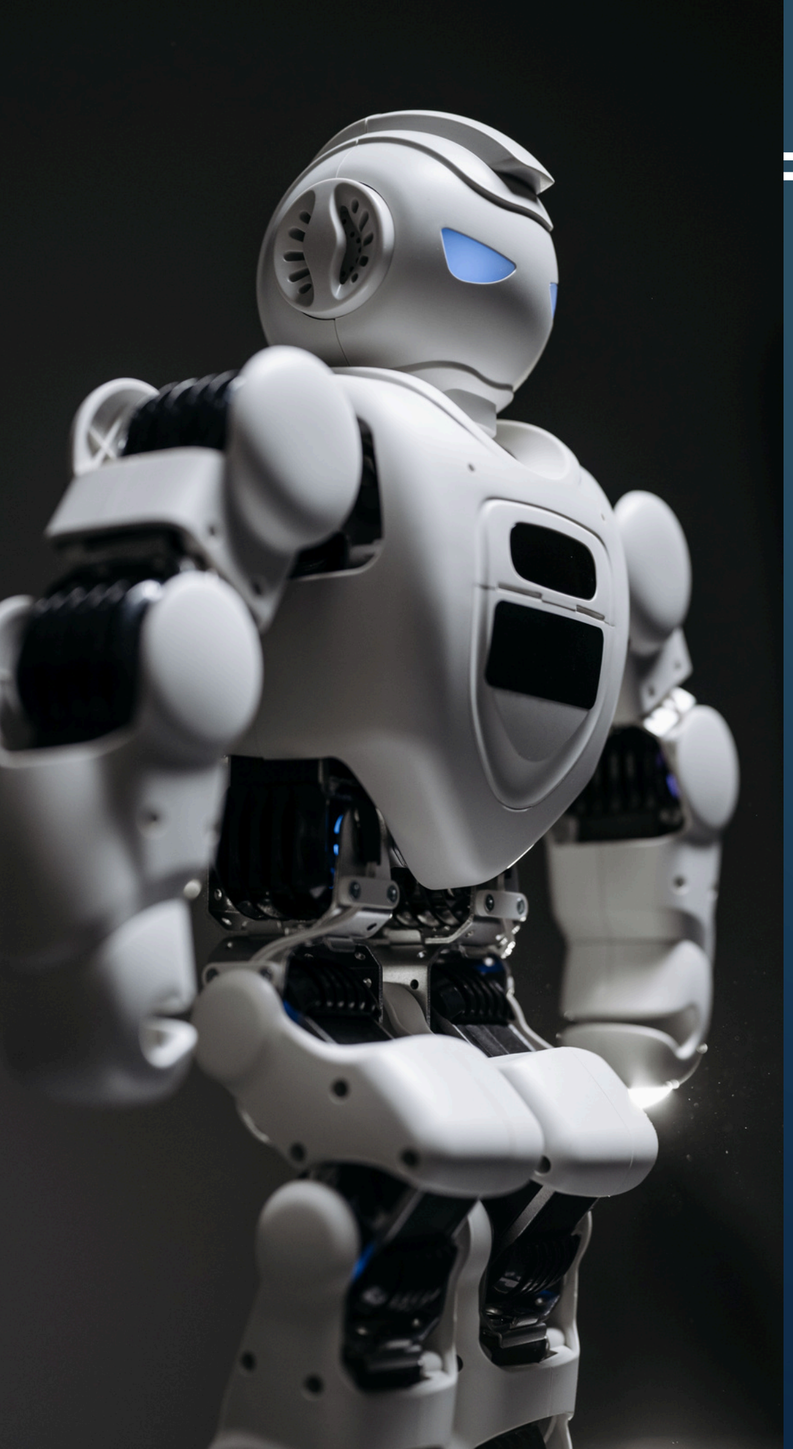
Electric Actuator



How ?

Receives a 50Hz PWM signal from an external controller, which it processes internally to determine the target position of its output shaft. Unlike raw DC motors, it relies on a specific timing protocol where the duration of a high-voltage pulse dictates the precise angle of rotation. We can use a library like the Arduino Servo.h library that implements this strict timing and background refresh protocol. Like DC motor the servo actuator can have stall current problem.

Specifications	Value
Operating Voltage	4.8V – 6.0V
No-Load Running Current	~200mA – 250mA
Stall Current (Peak)	~650mA – 750mA (at 5V)
Stall Torque	1.8 kg·cm (at 4.8V)
Operating Speed	0.12 seconds / 60° (at 4.8V)
Rotation Angle	~180° (90° in each direction)
Operating Temperature	0°C to 55°C



Eksperimen

Automated Gas Safety and Emergency Shut-Off System

```
void runRegularLoop() {
  int pirState = digitalRead(PIR_PIN);
  unsigned long currentMillis = millis();
  // If motion is detected and the servo isn't already open yet
  if (pirState == HIGH && !isOpen) {
    Serial.println("Motion detected! Opening servo...");
    myServo.write(SERVO_OPEN_ANGLE);
    isOpen = true;
    openStartTime = currentMillis;
  }
  // If the servo is open, check if the hold time has passed
  if (isOpen && (currentMillis - openStartTime >= HOLD_TIME)) {
    if (pirState == LOW) {
      Serial.println("Motion cleared. Closing servo...");
      myServo.write(SERVO_CLOSED_ANGLE);
      isOpen = false;
    } else {
      // If PIR is still HIGH, extend the timer window
      openStartTime = currentMillis;
    }
  }
  delay(50);
}
```

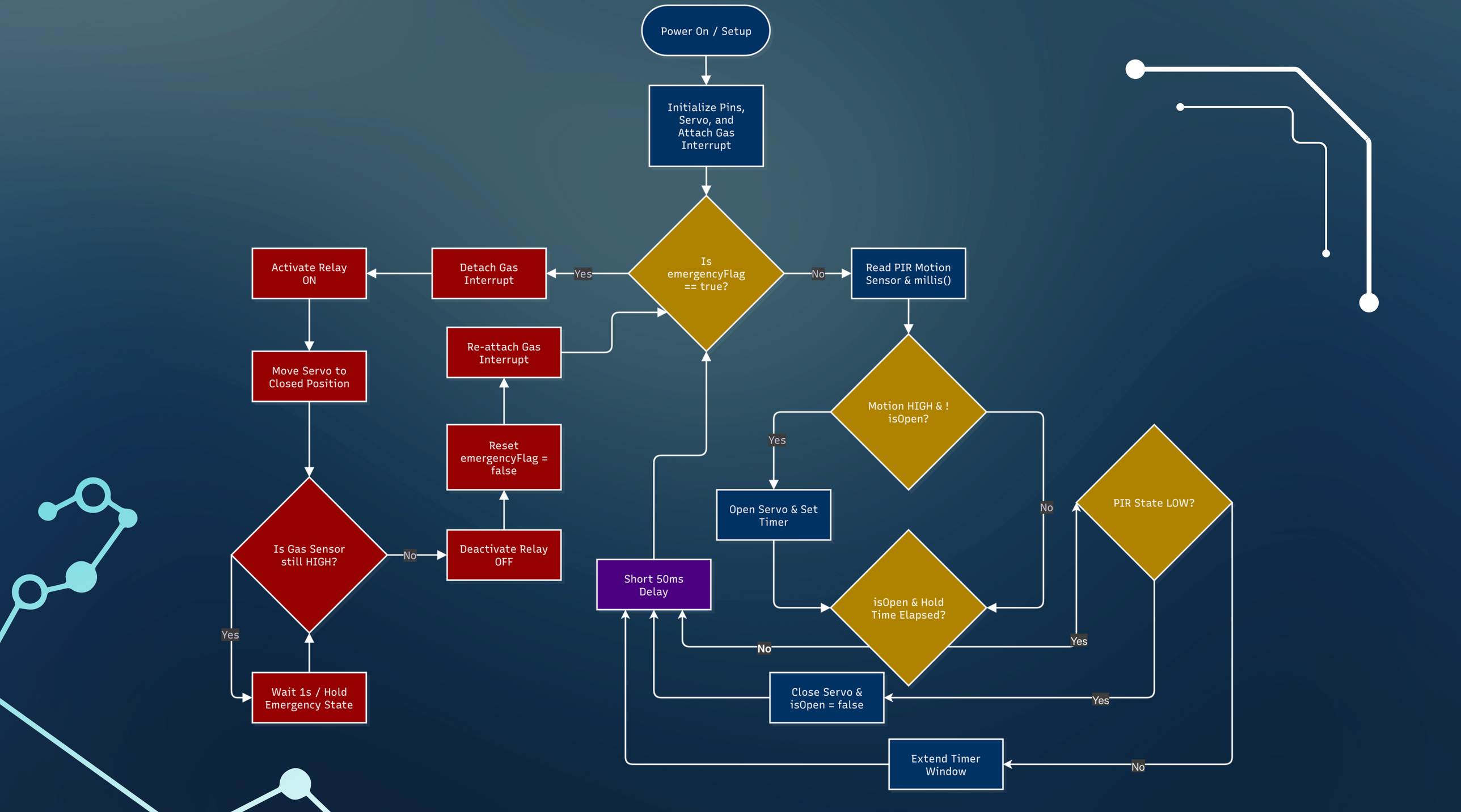
```
void gasDetectedISR() {
  if (digitalRead(GAS_PIN) == HIGH) {
    emergencyFlag = true;
  }
}
void runEmergencyLoop() {
  detachInterrupt(digitalPinToInterrupt(GAS_PIN));

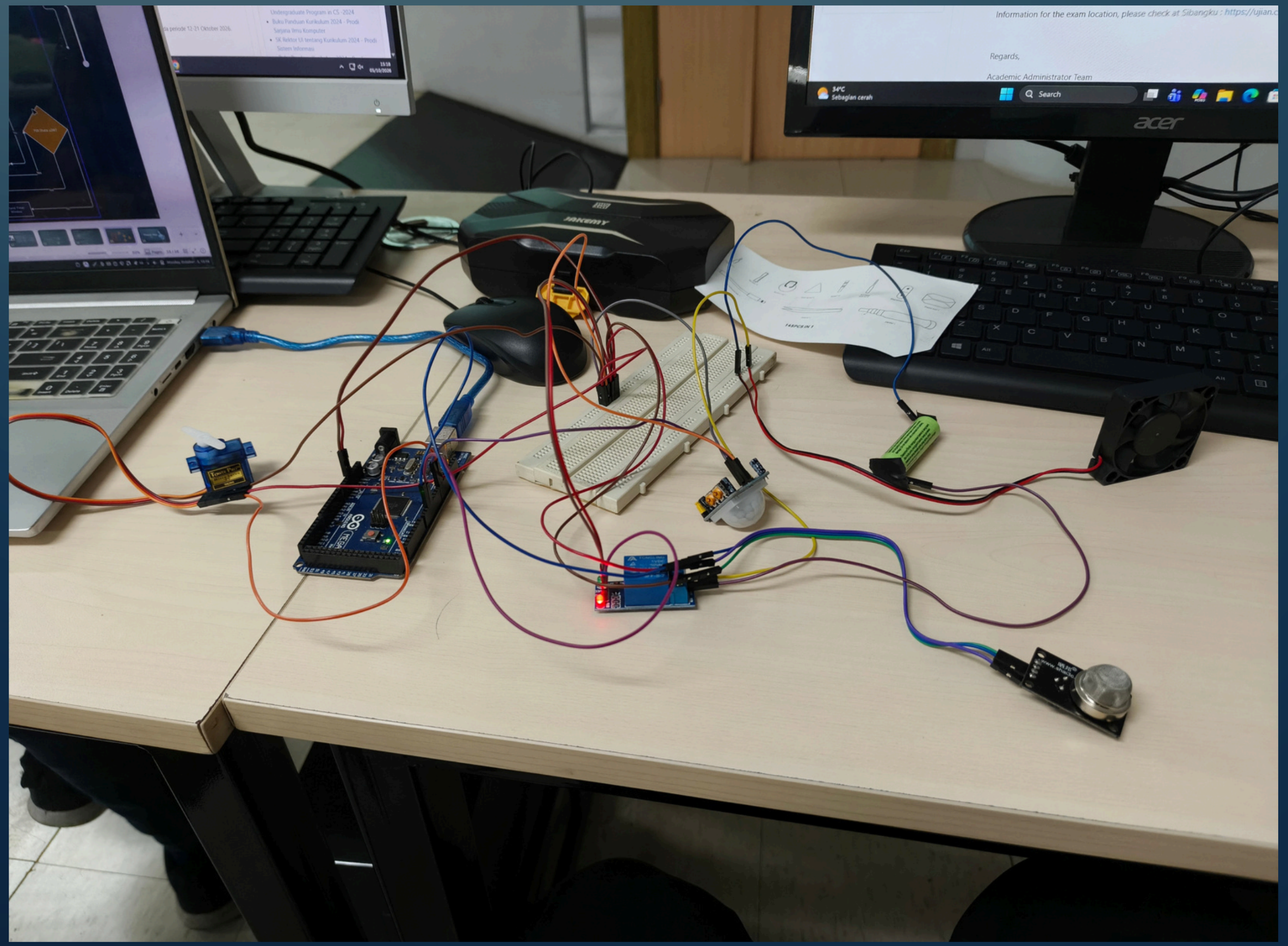
  Serial.println("EMERGENCY ALERT: Gas detected! Activating relay.");
  digitalWrite(RELAY_PIN, HIGH);
  myServo.write(SERVO_CLOSED_ANGLE);

  while (digitalRead(GAS_PIN) == HIGH) {
    Serial.println("WARNING: Gas levels still high. Holding emergency state...");
    delay(1000);
  }

  Serial.println("Gas cleared. Deactivating relay and resuming normal operation.");
  digitalWrite(RELAY_PIN, LOW);
  emergencyFlag = false;

  attachInterrupt(digitalPinToInterrupt(GAS_PIN), gasDetectedISR, RISING);
}
```


Power On / Setup
Initialize Pins, Servo, and Attach Gas Interrupt
Is emergencyFlag == true?
Yes
No
Detach Gas Interrupt
Activate Relay ON
Move Servo to Closed Position
Is Gas Sensor still HIGH?
Yes
No
Wait 1s / Hold Emergency State
Deactivate Relay OFF
Reset emergencyFlag = false
Re-attach Gas Interrupt
Read PIR Motion Sensor & millis()
Motion HIGH & ! isOpen?
Yes
No
Open Servo & Set Timer
isOpen & Hold Time Elapsed?
No
Yes
PIR State LOW?
Yes
No
Extend Timer Window
Close Servo & isOpen = false
Yes
Short 50ms Delay


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Thank You

